

Jianshu Zhang

Computer Science Ph.D. Student @ Northwestern University

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Research Summary

- I am a second-year **CS Ph.D.** student at **Northwestern University**. My research focuses on building **multimodal** and **agentic** models for the **interactive and physical world**, with a particular interest in **data-centric approaches** to evaluating and improving their capabilities.

Education

-  **Ph.D. in Computer Science** 2025-2030
Northwestern University Evanston, USA
Advisor: Prof. Han Liu · Cumulative GPA: 3.9/4.0
-  **B.Eng. in Information Security** 2021-2025
Wuhan University Wuhan, China
Rank: 1/182 · Outstanding Graduate

Research & Internship Experience

-  **Tencent AI Lab** Feb. 2025 - May. 2025
Mentor(s): Tianqing Fang
Project 1: WebCoT — Enhancing Web Agent Reasoning by Reconstructing CoT (EMNLP 2025 Findings)
Project 2: Explore to Evolve — Scaling Evolved Aggregation Logic for Deep Research Agents (ACL 2026 Main)
-  **MIT Media Lab** Oct. 2024 - Jan. 2025
Supervisor: Prof. Paul Liang, Dr. Yi R. (May) Fung
Project: VLM2-Bench — Evaluating How Well VLMs Implicitly Link Matching Visual Cues (ACL 2025 Main)
-  **Tong Zhang's Group, UIUC** Feb. 2024 - Dec. 2025
Supervisor: Prof. Tong Zhang
Project 1: Image Textualization — Automatic Framework for Rich & Detailed Image Descriptions (NeurIPS 2024)
Project 2: Personalized Visual Instruction Tuning — PVIT framework & PVIT-3M dataset (ICLR 2025)
Project 3: MLLM-Protector — Ensuring MLLM's Safety without Hurting Performance (EMNLP 2024)
Project 4: Bridge-Coder — Overcoming Language Gaps in Low-Resource Code (ACL 2025 Findings)

Publications

(◊ marks my first- or co-first-authored papers; * denotes equal contribution.)

- ◊ **PROGRESSLM: Towards Progress Reasoning in Vision-Language Models**
Jianshu Zhang*, Chengxuan Qian*, Haosen Sun, Haoran Lu, Dingcheng Wang, Letian Xue, Han Liu
ACL 2026 Main (Oral)
- ◊ **Progress Reward Modeling for Robotic Learning: A Comprehensive Survey**
Jianshu Zhang*, Keliang Wu*, Haoran Lu*, Anbang Liu, Ce Zhang, Weijie Yin, Chengxuan Qian, Xiyuan Yang, Zhenyu Pan, Guo Ye, Han Liu
Tech Report arXiv:2607.21655
- ◊ **SpaceNum: From Numerical Tokens to Visual Geometry**
Jianshu Zhang, Ce Zhang, Yijiang Li, Huifeixin Chen, Haoran Lu, Letian Xue, Bingyang Wang, Han Liu
Under Review 2026

- ◇ **Video2Skill: From Streaming Experience to Reusable Embodied Skills**
 Jianshu Zhang*, Ce Zhang*, Xiyuan Yang, Chenwei Xu, Haoran Lu, Yijiang Li, Yaqi Xie, Katia P. Sycara, Han Liu
Under Review 2026
- ◇ **ProgressCompass: Embodied Progress Reward Models Are Lost Without the Right Context**
 Jianshu Zhang*, Keliang Wu*, Chengxuan Qian, Xiyuan Yang, Haoran Lu, Anbang Liu, Han Liu
Under Review 2026
- ◇ **ManiMotion: Do Vision-Language Models Understand Fine-grained Manipulation Motion?**
 Yiming Liang*, Jianshu Zhang*, Ce Zhang*, Chenwei Xu, Weijie Yin, Han Liu
Under Review 2026
- **StreamScout: Learning When to Look Deeper for Streaming Video Understanding**
 Ce Zhang, Jing Bi, Jinxi He, Jianshu Zhang, Jingyang Lin, Yunzhong Xiao, Minghao Fu, Yaqi Xie, Katia P. Sycara, Ming Zhou
Under Review 2026
- **AnnotateAnything: Automatic Annotation of 3D Assets for Robot Manipulation**
 Haoran Lu, Mutian Shen, Shuyang Yu, Yu Xiao, Songling Liu, Jianshu Zhang, Shang Wu, Yue Chen, Guo Ye, Jiayi Wang, Zhaoran Wang, Han Liu
Under Review 2026
- **MagicSim: A Unified Infrastructure for Executable Embodied Interaction**
 Haoran Lu, Songling Liu, Yue Chen, Guo Ye, Mutian Shen, Shuyang Yu, Yu Xiao, Jihai Zhao, Shang Wu, Jianshu Zhang, Xiangtian Gui, Chuye Hong, Yuran Wang, Maojiang Su, Jiayi Wang, Ruihai Wu, Zhaoran Wang, Han Liu
Tech Report arXiv:2606.17511
- **MindCube: Spatial Mental Modeling from Limited Views**
 Qineng Wang, Baiqiao Yin, Pingyue Zhang, Jianshu Zhang, Kangrui Wang, Zihan Wang, Jieyu Zhang, Keshigeyan Chandrasegaran, Han Liu, Ranjay Krishna, Saining Xie, Jiajun Wu, Li Fei-Fei, Manling Li
ICLR 2026 (also SP4V@ICCV 2025 Best Paper Award)
- **Phys4D: Fine-Grained Physics-Consistent 4D Modeling from Video Diffusion**
 Haoran Lu, Shang Wu, Jianshu Zhang, Maojiang Su, Guo Ye, Chenwei Xu, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Zhaoran Wang, Han Liu
Under Review 2026
- **Explore to Evolve: Scaling Evolved Aggregation Logic via Proactive Online Exploration for Deep Research Agents**
 Rui Wang, Ce Zhang, Jun-Yu Ma, Jianshu Zhang, Hongru Wang, Yi Chen, Boyang Xue, Tianqing Fang, Zhisong Zhang, Hongming Zhang, Haitao Mi, Dong Yu, Kam-Fai Wong
ACL 2026 Main
- **AdvEvo-MARL: Shaping Internalized Safety through Adversarial Co-Evolution in Multi-Agent Reinforcement Learning**
 Zhenyu Pan, Yiting Zhang, Zhuo Liu, Yolo Yunlong Tang, Zeliang Zhang, Haozheng Luo, Yuwei Han, Jianshu Zhang, Dennis Wu, Hong-Yu Chen, Haoran Lu, Haoyang Fang, Manling Li, Chenliang Xu, Philip S. Yu, Han Liu
ICML 2026
- **VideoCritic: Diagnosing and Localizing Reasoning Errors in Video-Language Models**
 Chenwei Xu, Jianshu Zhang, Shang Wu, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Han Liu
VidLLMs@CVPR 2026

- ◇ **VLM2-Bench: A Closer Look at How Well VLMs Implicitly Link Explicit Matching Visual Cues**
 Jianshu Zhang*, Dongyu Yao*, Renjie Pi, Paul Pu Liang, Yi R. Fung
ACL 2025 Main
- **WebCoT: Enhancing Web Agent Reasoning by Reconstructing Chain-of-Thought in Reflection, Branching, and Rollback**
 Minda Hu, Tianqing Fang, Jianshu Zhang, Jun-Yu Ma, Zhisong Zhang, Jingyan Zhou, Hongming Zhang, Haitao Mi, Dong Yu, Irwin King
EMNLP 2025 Findings
- ◇ **Personalized Visual Instruction Tuning**
 Renjie Pi*, Jianshu Zhang*, Tianyang Han, Jipeng Zhang, Rui Pan, Tong Zhang
ICLR 2025
- ◇ **CAN: Leveraging Clients As Navigators for Generative Replay in Federated Continual Learning**
 Xuankun Rong*, Jianshu Zhang*, Kun He, Mang Ye
ICML 2025
- ◇ **Bridge-Coder: Transferring Model Capabilities from High-Resource to Low-Resource Programming Language**
 Jipeng Zhang*, Jianshu Zhang*, Yuanzhe Li*, Renjie Pi, Rui Pan, Runtao Liu, Ziqiang Zheng, Tong Zhang
ACL 2025 Findings
- **MultiVerse: A Multi-Turn Conversation Benchmark for Evaluating Large Vision and Language Models**
 Young-Jun Lee, Byung-Kwan Lee, Jianshu Zhang, Yechan Hwang, Byungsoo Ko, Han-Gyu Kim, Dongyu Yao, Xuankun Rong, Eojin Joo, Seung-Ho Han, Bowon Ko, Ho-Jin Choi
ICCV 2025
- **FairReason: Balancing Reasoning and Social Bias in MLLMs**
 Zhenyu Pan, Yutong Zhang, Jianshu Zhang, Haoran Lu, Haozheng Luo, Yuwei Han, Philip S. Yu, Manling Li, Han Liu
T2FM@ICCV 2025
- ◇ **Image Textualization: An Automatic Framework for Generating Rich and Detailed Image Descriptions**
 Renjie Pi*, Jianshu Zhang*, Jipeng Zhang, Rui Pan, Zhekai Chen, Tong Zhang
NeurIPS 2024
- ◇ **MLLM-Protector: Ensuring MLLM's Safety without Hurting Performance**
 Renjie Pi*, Tianyang Han*, Jianshu Zhang*, Yueqi Xie, Rui Pan, Qing Lian, Hanze Dong, Jipeng Zhang, Tong Zhang
EMNLP 2024 Main
- ◇ **FIRST: Teach a Reliable Large Language Model through Efficient Trustworthy Distillation**
 KaShun Shum*, Minrui Xu*, Jianshu Zhang*, Zixin Chen, Shizhe Diao, Hanze Dong, Jipeng Zhang, Muhammad Omer Raza
EMNLP 2024 Main
- ◇ **CORE: Mitigating Catastrophic Forgetting in Continual Learning through Cognitive Replay**
 Jianshu Zhang*, Yankai Fu*, Ziheng Peng*, Dongyu Yao, Kun He
CogSci 2024 (Oral)

◇ **FuzzLLM: A Novel and Universal Fuzzing Framework for Proactively Discovering Jailbreak Vulnerabilities in Large Language Models**

Dongyu Yao*, Jianshu Zhang*, Ian G. Harris, Marcel Carlsson

ICASSP 2024

Honors & Awards

- McCormick School of Engineering Fellowship, Northwestern University 2025
- National Scholarship (Top 0.2% nationally) 2024
- Lei Jun Computer Breakthrough Award (50K RMB), Wuhan University 2025
- SP4V@ICCV 2025 Best Paper Award (MindCube) 2025

Academic Service

- **Conference Reviewer:** ACL (Outstanding Review 2026), ICML (Silver Reviewer 2026), ICLR, NeurIPS, EMNLP, CVPR, ICCV, TMLR